

Prolog Program for PLANETS Database (AI)

Aim

To write a Prolog program to represent facts about the planets of the solar system and to query information such as position and presence of moons.

Objectives

- To represent facts of the form planet(Name, Position, HasMoon).
- To list all facts using a fail-driven backtracking loop.
- To use findall/3 to collect all planets that satisfy a condition.
- To query a planet based on its position in the solar system.

Algorithm

- 1 Start.
- 2 Define facts planet(Name, Position, HasMoon) for each of the eight planets.
- 3 Define a rule show_all to display every planet's details using backtracking.
- 4 Call show_all to list the entire planets database.
- 5 Use findall/3 to collect the list of all planets that have moons.
- 6 Query the database for the planet at a specific position.
- 7 Prolog unifies the query with the matching fact.
- 8 Display the retrieved planet name.
- 9 Stop.

Source Code (Prolog)

```
% Prolog program - PLANETS Database
planet(mercury, 1, no).
planet(venus, 2, no).
planet(earth, 3, yes).
planet(mars, 4, yes).
planet(jupiter, 5, yes).
planet(saturn, 6, yes).
planet(uranus, 7, yes).
planet(neptune, 8, yes).
show_all :-
    planet(Name, Position, HasMoon),
    format("~w -> Position: ~w, Has Moon(s): ~w~n", [Name, Position, HasMoon]),
    fail.
show_all.
:- write('Planets Database:'), nl,
   show_all,
   nl,
   findall(P, (planet(P, _, yes)), Moons),
   format("Planets with moons: ~w~n", [Moons]),
   planet(Name, 4, _),
   format("Query: planet at position 4 -> ~w~n", [Name]).
```

Output

```
Planets Database:
mercury -> Position: 1, Has Moon(s): no
venus -> Position: 2, Has Moon(s): no
earth -> Position: 3, Has Moon(s): yes
mars -> Position: 4, Has Moon(s): yes
jupiter -> Position: 5, Has Moon(s): yes
saturn -> Position: 6, Has Moon(s): yes
uranus -> Position: 7, Has Moon(s): yes
neptune -> Position: 8, Has Moon(s): yes
Planets with moons: [earth,mars,jupiter,saturn,uranus,neptune]
Query: planet at position 4 -> mars
```

Result

The Prolog program successfully created the Planets database, listed all planets with their positions and moon status, and correctly identified Mars as the planet at position 4.